

Environmental Studies – Part A Important Questions & Short Answers

1. Environmental Studies

Environmental Studies

Environmental Studies is the study of the relationship between humans and the environment. It includes natural resources, pollution, ecosystems, biodiversity, and methods of protecting nature for sustainable development.

2. Importance of EVS

Environmental Studies

- Creates awareness about environmental problems
 - Helps conserve natural resources
 - Promotes sustainable development
 - Encourages pollution control
 - Protects biodiversity and ecosystems
 - Improves human health and quality of life
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3. Land, Air and Marine Pollution

Land Pollution

Contamination of land due to waste disposal, plastics, pesticides, and industrial waste is called land pollution. It reduces soil fertility and affects living organisms.

Air Pollution

Air pollution is caused by smoke, harmful gases, vehicle emissions, and industries. It leads to respiratory diseases, global warming, and acid rain.

Marine Pollution

Pollution of oceans and seas due to oil spills, sewage, plastics, and industrial waste is called marine pollution. It harms aquatic life and marine ecosystems.

4. Disaster Management

Disaster Management

Disaster Management refers to planning and organizing resources to reduce the impact of natural and man-made disasters. It includes preparedness, response, recovery, and mitigation measures during floods, earthquakes, cyclones, fires, and other disasters.

5. Wildlife Protection Act

Wildlife Protection Act, 1972

The Wildlife Protection Act, 1972 is an Indian law enacted to protect wild animals, birds, and plants. It prevents illegal hunting and trade of wildlife and helps conserve biodiversity through national parks and wildlife sanctuaries.

6. Genetic Diversity

Genetic Diversity

Genetic diversity refers to the variation of genes within a species. It helps organisms adapt to environmental changes and increases the survival and evolution of species. High genetic diversity strengthens biodiversity and ecosystem stability.

7. Pond Ecosystem

Pond Ecosystem

A pond ecosystem is a freshwater ecosystem where plants, animals, and microorganisms interact with each other and the environment. Producers, consumers, and decomposers maintain ecological balance in the pond.

8. Solid Waste Management

Solid Waste Management

Solid Waste Management involves the collection, segregation, recycling, treatment, and disposal of waste materials. Proper waste management helps reduce pollution and protects public health.

9. Acid Rain & Soil Erosion

Acid Rain

Acid Rain

Acid rain occurs when sulfur dioxide and nitrogen oxides mix with rainwater. It damages crops, buildings, forests, and aquatic life.

Soil Erosion

Soil Erosion

Soil erosion is the removal of the top fertile layer of soil by wind, water, or human activities. It decreases soil fertility and agricultural productivity.

10. Human Population Growth

Human Population Growth

Human population growth refers to the increase in the number of people over time. Rapid population growth leads to pressure on natural resources, pollution, unemployment, and environmental degradation.

11. Industrial Pollution

Industrial Pollution

Industrial pollution is caused by the release of harmful gases, chemicals, and wastes from industries into air, water, and land. It affects human health and damages ecosystems.

12. Forest Conservation Act

Forest Conservation Act, 1980

The Forest Conservation Act, 1980 was enacted to protect forests and prevent deforestation in India. It regulates the use of forest land for non-forest purposes and promotes forest conservation.

13. Food Chain and Food Web

Food Chain

Food Chain

A food chain is a sequence in which energy passes from producers to consumers and decomposers through feeding relationships.

Food Web

Food Web

A food web is a network of interconnected food chains that shows multiple feeding relationships in an ecosystem.

14. Tribal Population and Right

Tribal Rights

Tribal people are indigenous communities that depend on forests and natural resources for their livelihood. They have rights related to land, forest resources, culture, education, and self-development protected by law.

15. Energy Resource

Energy Resources

Energy resources are sources used to produce power and energy. They are classified into renewable resources like solar and wind energy, and non-renewable resources like coal and petroleum.

16. India as a Mega Biodiversity

India Biodiversity

India is called a mega biodiversity country because it has a wide variety of plants, animals, forests, and ecosystems. The country contains many endemic and endangered species.

17. Effects and Controls of Air Pollution

Air Pollution

Effects

- Respiratory diseases
- Global warming
- Acid rain
- Damage to plants and buildings

Control Measures

- Use public transport
 - Plant more trees
 - Reduce industrial emissions
 - Use clean and renewable energy sources
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18. Effects of Noise Pollution

Noise Pollution

Noise pollution causes stress, hearing loss, high blood pressure, sleep disturbance, and reduced concentration. It is mainly caused by vehicles, industries, and loudspeakers.

19. Alternate Energy Sources

Renewable Energy

Alternate energy sources are non-conventional and renewable sources of energy such as solar energy, wind energy, tidal energy, geothermal energy, and biogas. They reduce environmental pollution and conserve fossil fuels.

20. Species Diversity

Species Diversity

Species diversity refers to the variety and abundance of different species present in an ecosystem. Greater species diversity increases ecosystem stability and ecological balance.

Part-B Answers

1. Nature, Scope & Importance of EVS

Nature of EVS

Environmental Studies (EVS) is multidisciplinary in nature because it combines subjects like biology, chemistry, geography, economics, sociology, and political science to study the environment and human interactions with it. It focuses on protecting natural resources and maintaining ecological balance.

Scope of EVS

The scope of EVS is very wide and includes:

- Study of natural resources like water, forest, minerals, and energy
- Ecosystem and biodiversity conservation
- Environmental pollution and its control
- Social issues related to the environment
- Sustainable development
- Human population and environment relationship

Importance of EVS

- Creates awareness about environmental problems
 - Helps in conservation of natural resources
 - Encourages pollution control
 - Promotes sustainable development
 - Protects biodiversity and ecosystems
 - Improves human health and quality of life
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2. Deforestation

Meaning

Deforestation refers to the large-scale removal of forests for agriculture, industries, urbanization, and other human activities.

Causes of Deforestation

Agricultural Expansion

Forests are cleared for farming and cultivation.

Urbanization

Construction of roads, buildings, and cities leads to forest destruction.

Industrialization

Industries require land and forest products.

Mining Activities

Mining destroys vegetation and forest cover.

Overgrazing

Excessive grazing by animals damages forest regeneration.

Effects of Deforestation

Soil Erosion

Removal of trees exposes soil to wind and rain.

Loss of Biodiversity

Many plants and animals lose their habitat.

Global Warming

Reduction of trees increases carbon dioxide in the atmosphere.

Disturbance of Water Cycle

Rainfall patterns get affected.

Floods and Droughts

Environmental imbalance causes natural disasters.

Control Measures

- Afforestation
 - Reforestation
 - Social forestry
 - Strict forest laws
 - Public awareness programs
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3. In-situ & Ex-situ Conservation of Biodiversity

In-situ Conservation

Meaning

Protection of plants and animals in their natural habitats.

Methods

National Parks

Protected areas for wildlife and plants.

Wildlife Sanctuaries

Areas where animals are protected from hunting.

Biosphere Reserves

Large protected areas preserving biodiversity.

Advantages

- Maintains natural ecosystem
 - Species evolve naturally
 - Conserves many species together
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Ex-situ Conservation

Meaning

Conservation of organisms outside their natural habitats.

Methods

Zoos

Protection and breeding of animals.

Botanical Gardens

Conservation of rare plants.

Seed Banks

Storage of seeds for future use.

Gene Banks

Storage of genetic material.

Advantages

- Protects endangered species
 - Scientific breeding possible
 - Useful during natural disasters
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4. Environmental Laws & Its Importance

Meaning

Environmental laws are legal measures created to protect the environment from damage and pollution.

Important Environmental Laws in India

Environment Protection Act, 1986

Provides overall protection to the environment.

Water (Prevention and Control of Pollution) Act, 1974

Controls water pollution.

Air (Prevention and Control of Pollution) Act, 1981

Controls air pollution.

Wildlife Protection Act, 1972

Protects wild animals and birds.

Forest Conservation Act, 1980

Prevents unnecessary deforestation.

Importance of Environmental Laws

- Prevent pollution
 - Protect forests and wildlife
 - Ensure public health safety
 - Promote sustainable development
 - Maintain ecological balance
-

5. Water Resources

Meaning

Water resources are natural sources of water available for human use.

Types of Water Resources

Surface Water

Water found in rivers, lakes, ponds, and reservoirs.

Groundwater

Water stored beneath the earth's surface.

Rainwater

Water obtained directly from rainfall.

Uses of Water Resources

- Drinking and domestic use
- Irrigation in agriculture
- Industrial purposes
- Hydroelectric power generation

Problems Related to Water Resources

Water Scarcity

Insufficient water availability.

Water Pollution

Contamination by sewage and chemicals.

Overuse of Groundwater

Excess extraction lowers water table.

Conservation Methods

- Rainwater harvesting
 - Water recycling
 - Afforestation
 - Efficient irrigation systems
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6. Role of Nuclear Hazards in Human Health Risks

Meaning

Nuclear hazards arise from radioactive substances that emit harmful radiation.

Sources of Nuclear Hazards

- Nuclear power plants
- Nuclear weapons testing
- Medical radiation exposure
- Radioactive waste

Health Risks

Cancer

Radiation damages body cells.

Genetic Mutations

Changes occur in genes and chromosomes.

Birth Defects

Radiation affects unborn children.

Skin Diseases

Exposure causes burns and skin problems.

Organ Damage

Internal organs may be affected.

Preventive Measures

- Proper disposal of radioactive waste
 - Safety measures in nuclear plants
 - Radiation monitoring
 - Use of protective equipment
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7. Energy Resources

Meaning

Energy resources are materials or natural phenomena used to produce power.

Types of Energy Resources

Non-renewable Energy Resources

These cannot be replaced quickly.

Coal

Used in thermal power plants.

Petroleum

Used as fuel in vehicles.

Natural Gas

Used for cooking and industries.

Renewable Energy Resources

These can be naturally replenished.

Solar Energy

Energy obtained from sunlight.

Wind Energy

Energy produced using wind turbines.

Hydropower

Electricity generated from flowing water.

Biomass Energy

Energy obtained from organic matter.

Importance of Energy Resources

- Supports industries
- Helps transportation
- Produces electricity
- Improves economic development

Need for Conservation

- Fossil fuels are limited
- Reduces pollution
- Ensures energy availability for future generations

8. Grassland Ecosystem

Meaning

A grassland ecosystem is an area where grasses dominate and trees are very few.

Types of Grasslands

Tropical Grasslands

Found in warm regions with seasonal rainfall.

Temperate Grasslands

Found in moderate climate regions.

Components of Grassland Ecosystem

Abiotic Components

- Soil
- Sunlight
- Water
- Temperature

Biotic Components

Producers

Grasses and small plants.

Consumers

Herbivores like deer and carnivores like lions.

Decomposers

Bacteria and fungi decompose dead matter.

Importance of Grassland Ecosystem

- Provides grazing land
- Prevents soil erosion
- Maintains biodiversity
- Supports food chains

9. Environmental Laws and Its Importance

Objectives of Environmental Laws

- Prevent environmental pollution

- Protect wildlife and forests
- Conserve natural resources
- Ensure sustainable development

Importance

Pollution Control

Reduces air, water, and soil pollution.

Public Health Protection

Protects humans from environmental hazards.

Biodiversity Conservation

Preserves plants and animals.

Sustainable Use of Resources

Encourages responsible use of natural resources.

10. Renewable Resources

Meaning

Renewable resources are resources that can naturally regenerate after use.

Types of Renewable Resources

Solar Energy

Energy from the sun.

Wind Energy

Energy generated from wind movement.

Hydroelectric Energy

Energy from flowing water.

Biomass

Energy from organic materials.

Geothermal Energy

Energy from heat inside the earth.

Advantages

- Eco-friendly
- Reduces pollution
- Unlimited availability
- Sustainable source of energy

Disadvantages

- High installation cost
- Depends on climate conditions
- Requires advanced technology

Importance

- Reduces dependence on fossil fuels
- Controls global warming
- Promotes sustainable development

11. Environmental Protection Acts

Meaning

Environmental Protection Acts are laws enacted by the government to protect the environment and control pollution. These laws help in conserving natural resources and maintaining ecological balance.

Major Environmental Protection Acts in India

1. Environment Protection Act, 1986

Meaning

This act was passed after the Bhopal Gas Tragedy to provide overall protection to the environment.

Objectives

- Protect and improve environmental quality
- Control pollution
- Regulate industrial activities

Important Features

- Government can set pollution standards
- Restricts harmful industrial activities
- Punishment for environmental violations

Importance

- Acts as an umbrella law for environmental protection
 - Helps control pollution effectively
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2. Water (Prevention and Control of Pollution) Act, 1974

Objectives

- Prevent water pollution
- Maintain water quality

Features

- Establishment of Pollution Control Boards
- Industries must treat wastewater before discharge

Importance

- Protects rivers, lakes, and groundwater
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3. Air (Prevention and Control of Pollution) Act, 1981

Objectives

- Prevent and control air pollution

Features

- Controls industrial emissions
- Sets air quality standards

Importance

- Reduces harmful gases and smoke
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4. Wildlife Protection Act, 1972

Objectives

- Protect wild animals and birds
- Prevent illegal hunting

Features

- Creation of national parks and sanctuaries
- Protection for endangered species

Importance

- Conserves biodiversity
-

5. Forest Conservation Act, 1980

Objectives

- Prevent deforestation
- Conserve forest resources

Features

- Restricts conversion of forest land for non-forest use

Importance

- Protects ecological balance
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Importance of Environmental Protection Acts

- Protect natural resources
- Control environmental pollution

- Protect human health
 - Preserve biodiversity
 - Promote sustainable development
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12. Biodiversity at Global Level

Meaning

Biodiversity refers to the variety of living organisms present on Earth including plants, animals, and microorganisms.

Levels of Biodiversity

1. Genetic Diversity

Variation in genes within a species.

Example

Different varieties of rice and mango.

2. Species Diversity

Variety of species present in a region.

Example

Different animals and plants in forests.

3. Ecosystem Diversity

Variety of ecosystems such as forests, deserts, and oceans.

Global Distribution of Biodiversity

Tropical Rainforests

Contain the highest biodiversity due to favorable climate.

Coral Reefs

Known as the “rainforests of the sea.”

India as a Mega-diversity Nation

India has rich biodiversity with many endemic species.

Threats to Global Biodiversity

- Deforestation
 - Pollution
 - Climate change
 - Overexploitation
 - Habitat destruction
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Conservation of Global Biodiversity

In-situ Conservation

Protection within natural habitats.

Ex-situ Conservation

Protection outside natural habitats.

International Efforts

- Convention on Biological Diversity (CBD)
 - UNESCO Biosphere Programme
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Importance of Biodiversity

- Maintains ecological balance
- Provides food and medicine

- Supports ecosystem services
 - Preserves natural beauty
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13. Methods of Controlling Water Pollution

Meaning

Water pollution control refers to methods used to reduce contamination in water bodies.

Methods of Controlling Water Pollution

1. Treatment of Sewage

Primary Treatment

Removal of solid waste.

Secondary Treatment

Biological treatment using microorganisms.

Tertiary Treatment

Chemical treatment for purification.

2. Industrial Waste Treatment

Industries should treat harmful chemicals before releasing wastewater.

3. Proper Disposal of Waste

Avoid dumping plastics and garbage into water bodies.

4. Use of Eco-friendly Products

Reduce use of harmful detergents and chemicals.

5. Rainwater Harvesting

Improves groundwater recharge and reduces water scarcity.

6. Afforestation

Trees help maintain the water cycle and prevent soil erosion.

7. Public Awareness

Educating people about water conservation and pollution prevention.

Importance of Controlling Water Pollution

- Protects aquatic life
 - Provides safe drinking water
 - Prevents diseases
 - Maintains ecosystem balance
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14. Causes, Effects and Control of Air Pollution

Air Pollution

Meaning

Air pollution is the contamination of air by harmful gases, smoke, dust, and chemicals.

Causes of Air Pollution

1. Vehicle Emissions

Release carbon monoxide, nitrogen oxides, and smoke.

2. Industrial Emissions

Factories release harmful gases and particulate matter.

3. Burning of Fossil Fuels

Coal and petroleum release pollutants during combustion.

4. Deforestation

Reduces oxygen and increases carbon dioxide.

5. Burning of Waste

Burning plastics and garbage produces toxic gases.

Effects of Air Pollution

1. Human Health Problems

- Asthma
- Bronchitis
- Lung cancer
- Respiratory diseases

2. Global Warming

Increase in greenhouse gases raises Earth's temperature.

3. Acid Rain

Sulphur dioxide and nitrogen oxides mix with rainwater.

4. Damage to Plants

Affects photosynthesis and crop growth.

5. Ozone Layer Depletion

Harmful UV rays enter Earth's atmosphere.

Control Measures of Air Pollution

1. Use of Public Transport

Reduces vehicle emissions.

2. Afforestation

Trees absorb carbon dioxide.

3. Use of Clean Energy

Solar and wind energy reduce pollution.

4. Pollution Control Devices

Industries use filters and scrubbers.

5. Avoid Burning Waste

Proper waste management reduces toxic gases.

6. Environmental Laws

Government regulations help control emissions.

15. Structure and Function of Ecosystem

Ecosystem

Meaning

An ecosystem is a community of living organisms interacting with each other and with the physical environment.

Structure of Ecosystem

The ecosystem has two main components:

1. Abiotic Components

Non-living components of the environment.

Examples

- Air
 - Water
 - Soil
 - Temperature
 - Sunlight
-

2. Biotic Components

Living organisms in the ecosystem.

Producers

Green plants that prepare food through photosynthesis.

Consumers

Animals that depend on plants or other animals for food.

Types of Consumers

- Primary consumers (herbivores)
- Secondary consumers (carnivores)
- Tertiary consumers

Decomposers

Bacteria and fungi that decompose dead organisms.

Functions of Ecosystem

1. Energy Flow

Energy flows from producers to consumers through food chains.

2. Nutrient Cycling

Recycling of nutrients between living organisms and the environment.

3. Food Chain and Food Web

Transfer of food and energy among organisms.

4. Ecological Balance

Maintains stability in nature.

5. Decomposition

Breakdown of dead organic matter by decomposers.

Importance of Ecosystem

- Supports life on Earth
- Maintains environmental balance
- Provides food and shelter
- Helps nutrient recycling
- Regulates climate and water cycle